F18CF – Linear Algebra

Tutorial Sheet 3: Vector Spaces

Getting started.

1. We consider the vector space $\mathcal{P}_2(\mathbb{R})$ and introduce the following polynomial functions $P_1, P_2 \in \mathcal{P}_2(\mathbb{R})$:
$$P_1(x) := x^2 + x + 1, \quad P_2(x) := x^2 - 3x - 1.$$
Determine whether the following polynomial functions are linear combinations of P_1 and P_2:
   (a) $x^2 + 3x + 2$.
   (b) x^2.

2. We consider the functions $f, g \in \mathcal{F}(\mathbb{R})$ defined by:
$$f(x) := \cos(x), \quad g(x) := \sin(x).$$
   (a) Show that f, g are linearly independent. (Hint: Assume that there exists a linear combination of the form $\lambda f(x) + \mu g(x) = 0$ for every $x \in \mathbb{R}$, and evaluate this equation at $x = 0$ and $x = \pi/2$.)
   (b) Show that the function $\cos(2x)$ is not a linear combination of f and g. (Hint: Assume that there exists a linear combination of the form $\lambda f(x)+\mu g(x) = \cos(2x)$ for every $x \in \mathbb{R}$, and evaluate this equation at well chosen values of x.)

3. We consider the space $\mathcal{P}_3(\mathbb{R})$ with its basis $\mathcal{B} = (1, x, x^2, x^3)$. We consider the following polynomial functions $P_1, \ldots, P_4$ defined by:
$$P_1(x) = x^3, \quad P_2(x) = x^2(x-1), \quad P_3(x) = x(x-1)^2, \quad P_4(x) = (x-1)^3.$$
   (a) Write down the vector $[P_i]_{\mathcal{B}} \in \mathbb{R}^4$ for each $1 \leq i \leq 4$.
   (b) Show that $P_1, \ldots, P_4$ is a basis of $\mathcal{P}_3(\mathbb{R})$.

Problems.

4. **Polynomial interpolation revisited.** Let $x_1, \ldots, x_n \in \mathbb{R}$ be pairwise distinct, and let $y_1, \ldots, y_n \in \mathbb{R}$. We would like to find a polynomial function P such that $P(x_i) = y_i$ for every i.

   We introduce the following *Lagrange polynomials* $L_1, \ldots, L_n$ defined as follows. For $1 \leq k \leq n$, we set:
$$L_k(x) := \prod_{i=1, i\neq k}^{n} (x - x_i).$$
   For instance, $L_1(x) = (x - x_2) \times (x - x_3) \cdots (x - x_n)$, $\quad L_2(x) = (x - x_1) \times (x - x_3) \times (x - x_4) \cdot \cdots \cdot (x - x_n)$, etc.

(a) Let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$ be such that $\alpha_1 L_1(x) + \ldots + \alpha_n L_n(x) = 0$ for every $x \in \mathbb{R}$. By evaluating this equality at judiciously chosen points, show that $\alpha_1 = \ldots = \alpha_n = 0$. What does that imply for the family of polynomials $L_1, \ldots, L_n$?

(b) Show that $L_1, \ldots, L_n$ is a basis of $\mathcal{P}_{n-1}(\mathbb{R})$.

(c) Show that there exists a unique polynomial P of degree at most $n-1$, called the *interpolating polynomial*, such that

$$P(x_1) = y_1, \ldots, P(x_n) = y_n.$$

(d) Find the interpolating polynomial satisfying the equations:

$$P(1) = 1, \quad P(2) = 8, \quad P(3) = 27.$$

5. **Dimension of the space of real sequences.** We consider the space $\mathbb{R}^{\mathbb{N}}$ of real sequences. For every $k \geq 0$, we define the following sequence

$$\delta_k := \begin{pmatrix} \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$$

such that the k-th term is 1, and all the other terms are zero.

(a) Show that for every k, the family $\delta_0, \ldots, \delta_k$ is linearly independent.

(b) Deduce that $\dim \mathbb{R}^{\mathbb{N}} = \infty$.

6. **A linearly independent family of exponential functions.** We want to show that the functions e^x, e^{2x}, and e^{4x} form a linearly independent family of vectors of the space $\mathcal{F}(\mathbb{R})$ of real functions. Let $a, b, c \in \mathbb{R}$ such that

$$ae^x + be^{2x} + ce^{4x} = 0 \quad \text{for every } x \in \mathbb{R}.$$

(a) By evaluating this equation at a judiciously chosen point, show that $a+b+c = 0$.

(b) By differentiating the equation $ae^x + be^{2x} + ce^{4x} = 0$, show that we also have $a + 2b + 4c = 0$ and $a + 4b + 16c = 0$.

(c) Show that $a = b = c = 0$ and conclude that these functions are linearly independent.

7. **Using coordinate vectors.** The goal of this exercise is to prove some of the results mentioned in the course. Let V be a n-dimensional vector space with basis $\mathcal{B}$, and let $\vec{v}_1, \ldots, \vec{v}_k$ be a family of vectors of V.

(a) Show that if $[\vec{v}_1]_{\mathcal{B}}, \ldots, [\vec{v}_k]_{\mathcal{B}}$ is linearly independent, then so is $\vec{v}_1, \ldots, \vec{v}_k$.

(b) Show that if $[\vec{v}_1]_{\mathcal{B}}, \ldots, [\vec{v}_k]_{\mathcal{B}}$ is spanning, then so is $\vec{v}_1, \ldots, \vec{v}_k$.